

Business Insights & Marketing Recommendations Report

Module 8 — From Customer Segmentation to Marketing Strategy

Prepared for: Machine Learning Team Leader

Dataset: Customer Personality / Marketing Campaign data (2,240 customers)

Segmentation method: K-Means clustering ($k = 4$), standardized features

Contents: Segment Review · Personas · Business Opportunities · Marketing Strategy · Product Recommendations · Pricing & Discounts · Retention · Reactivation · Campaign Action Plan · Strategy Matrix · Dashboard · Executive Summary

Project Overview

This report translates the customer segmentation produced in the prior module into concrete marketing and business actions. Using the engineered feature set (age, income, total spend, recency, tenure, family composition, purchase frequency, campaign response, and channel activity), a K-Means model was fit with $k = 4$ clusters, chosen to balance statistical separation with a business-usable number of actionable segments. Each segment is profiled, named, and translated into a persona, an opportunity assessment, and a full set of marketing, pricing, retention, and reactivation recommendations.

Task 1 — Customer Segment Summary

Four segments were identified. Ranked by average customer value, Premium Loyal Customers and Affluent Family Shoppers together represent under half of the base by headcount but drive the large majority of revenue (see Task 11 dashboard).

Segment	Business Name	Customers	Percentage
Cluster 2	Premium Loyal Customers	498	22.2%
Cluster 3	Affluent Family Shoppers	605	27.0%
Cluster 0	Budget-Conscious Families	396	17.7%
Cluster 1	Low-Engagement Shoppers	741	33.1%

Task 2 — Customer Personas

Premium Loyal Customers (22.2% · 498 customers)

Attribute	Detail
Income Level	High — average income ~\$78,000, top tier of the customer base
Spending Behavior	Very high annual spend (~\$1,430/yr), concentrated in wine and meat products
Purchase Frequency	Highest of all segments (~19 purchases per period)
Preferred Products	Wine, meat, and premium/gold items
Campaign Response	Highest acceptance rate (~20%) — most responsive to offers
Shopping Channel	Balanced across catalog, store, and web; catalog usage far above other segments
Deal Dependency	Low — purchases driven by preference and quality, not discounting
Customer Activity	Highly active; converts web visits into purchases efficiently
Risk Level	Very low churn risk
Lifetime Value	Highest lifetime value in the base

Affluent Family Shoppers (27.0% · 605 customers)

Attribute	Detail
Income Level	High-moderate — average income ~\$60,500
Spending Behavior	High annual spend (~\$830/yr), second only to Premium Loyal Customers
Purchase Frequency	High (~18.5 purchases per period)
Preferred Products	Wine and meat lead, balanced basket across categories, family-sized baskets
Campaign Response	Moderate acceptance rate (~5%)
Shopping Channel	Strong web and store activity; highest average web purchases of any segment
Deal Dependency	Moderate-to-high — the most deal-responsive high-spending segment
Customer Activity	Very active across all channels; larger households with teens at home
Risk Level	Low churn risk
Lifetime Value	High — second-highest lifetime value

Budget-Conscious Families (17.7% · 396 customers)

Attribute	Detail
Income Level	Low-moderate — average income ~\$41,600
Spending Behavior	Low annual spend (~\$137/yr)
Purchase Frequency	Low (~6.6 purchases per period)
Preferred Products	Wine over-indexes relative to other categories, but overall basket is small
Campaign Response	Very low acceptance rate (~3%)
Shopping Channel	High web visits (6.3/month) but low conversion — mostly browsing

Attribute	Detail
Deal Dependency	High — deals are the largest share of their purchase activity
Customer Activity	Larger families (avg. 1.1 kids, 1.0 teens at home); oldest average age (~61)
Risk Level	Moderate-to-high churn risk — price sensitive and budget constrained
Lifetime Value	Low-to-moderate lifetime value

Low-Engagement Shoppers (33.1% · 741 customers)

Attribute	Detail
Income Level	Lowest — average income ~\$33,800
Spending Behavior	Lowest annual spend (~\$120/yr)
Purchase Frequency	Lowest (~6.2 purchases per period)
Preferred Products	No strong category preference; minimal spend across the board
Campaign Response	Lowest acceptance rate (~3%)
Shopping Channel	Lowest purchases across every channel; highest browse-to-buy ratio (visits without converting)
Deal Dependency	Low in absolute terms, but price sensitivity is implied by low spend at any price point
Customer Activity	Largest segment by headcount (33% of the base); smaller households, comparatively younger
Risk Level	Highest disengagement / churn risk
Lifetime Value	Lowest lifetime value per customer, but largest volume — a scale opportunity if activated

Task 3 — Business Opportunity Analysis

Premium Loyal Customers

Question	Answer
What makes this segment valuable?	Highest income, highest spend, and highest campaign response — they generate outsized revenue per customer and respond to offers others ignore.
Business opportunity	Upsell premium/limited editions, deepen wallet share via catalog and loyalty perks, use as brand advocates and referral source.
Business risk	Complacency risk — competitors targeting high-value customers with better loyalty perks; over-contacting could fatigue the relationship.
Marketing objective	Protect and grow share of wallet; convert them into referral and advocacy channels.

Affluent Family Shoppers

Question	Answer
What makes this segment valuable?	High spend combined with family purchasing patterns and above-average deal responsiveness — a large, high-value, still price-aware group.
Business opportunity	Family bundles, multi-buy promotions, cross-sell into new categories; second-largest revenue pool with room to grow frequency.
Business risk	Deal-sensitivity could erode margin if discounting is not tightly targeted; family budgets are more elastic than Premium Loyal customers.
Marketing objective	Increase purchase frequency and basket size through targeted bundles without over-discounting.

Budget-Conscious Families

Question	Answer
What makes this segment valuable?	Large households with real purchase intent (high web visits, deal usage) but a spending ceiling — value lies in efficient, low-cost engagement.
Business opportunity	Convert browsing into buying with well-timed, low-cost incentives (free shipping, small deals); grow small basket sizes over time.
Business risk	Highest sensitivity to price change; risk of switching to lower-cost competitors and of low response to non-discount campaigns.
Marketing objective	Improve conversion of existing engagement (visits) into completed purchases.

Low-Engagement Shoppers

Question	Answer
What makes this segment valuable?	The largest segment by headcount (33%) — low current value per customer, but scale means even modest activation moves the needle at portfolio level.
Business opportunity	Reactivation and awareness campaigns; identify a subset with reactivation potential vs. truly disengaged.

Question	Answer
Business risk	Highest churn/inactivity risk; without intervention this segment is the most likely source of long-term attrition.
Marketing objective	Re-engage a meaningful share of this group and prevent silent churn.

Task 4 — Marketing Strategy Design

Segment	Message	Tone	Personalized Offer	Channel	Timing	Frequency
Premium Loyal Customers	Exclusive access to our finest selections, first	Refined, appreciative, exclusive	Early access to new/limited releases + VIP loyalty tier	Email + Catalog	Product launches / holidays	Weekly
Affluent Family Shoppers	More of what your family loves, together	Warm, family-oriented	Family bundle discounts, multi-buy offers	Email + Web push	Weekends / school holidays	Bi-weekly
Budget-Conscious Families	Save more on the essentials you already buy	Practical, reassuring, value-focused	Percentage-off coupons, free shipping thresholds	SMS + Email	Paydays / start of month	Bi-weekly
Low-Engagement Shoppers	We miss you — here's something to bring you back	Friendly, low-pressure, inviting	Welcome-back discount or free gift with purchase	Email + Retargeting ads	Anytime (trigger-based on inactivity)	Monthly

Task 5 — Product Recommendation Strategy

Segment	Primary Category	Secondary Category	Cross-Sell	Upsell	Bundle Idea
Premium Loyal Customers	Wine	Meat	Gourmet cheese / fish pairings	Limited-edition & reserve wines	Premium tasting bundle (wine + meat + gold items)
Affluent Family Shoppers	Wine	Meat	Sweets & fruits for family occasions	Larger family-size packs	Family dinner bundle (meat + wine + sides)
Budget-Conscious Families	Wine (value tier)	Gold/novelty items	Everyday staples at entry price points	Mid-tier product lines	Value multi-pack bundle
Low-Engagement Shoppers	Entry-level / best-seller items	Gold/novelty items	Low-cost trial-size products	Starter bundles to build habit	Welcome-back sampler bundle

Task 6 — Pricing and Discount Strategy

Segment	Discount %	Coupon Strategy	Promotional Offers	Seasonal Offers	Premium Pricing Opportunity
Premium Loyal Customers	0-5% (rarely discounted)	Invite-only, exclusivity-framed coupons	Early access instead of price cuts	Holiday gift sets at full/near-full price	High — willing to pay for exclusivity and quality
Affluent Family Shoppers	10-15%	Multi-buy / family bundle coupons	'Buy more, save more' tiers	Back-to-school, holiday family bundles	Moderate — premium lines with family framing
Budget-Conscious Families	15-20%	Frequent, easily redeemable coupons	Free shipping thresholds, small basket discounts	Start-of-month / payday promotions	Low — focus on value, not premium upsell
Low-Engagement Shoppers	20-25% (win-back)	Single-use, time-limited win-back coupons	'Welcome back' first-purchase incentive	Re-permission campaigns at year-end/New Year	Low — priority is reactivation over margin

Task 7 — Customer Retention Strategy

Segment	Loyalty Rewards	Retention Campaign	Follow-up Frequency	Support Priority	Membership Plan	Referral Incentive
Premium Loyal Customers	Tiered VIP loyalty program with points on every purchase	Personal thank-you + early access to new arrivals	Monthly check-in	Highest (dedicated support line)	Invite-only premium membership	High-value referral credit
Affluent Family Shoppers	Family points program with milestone rewards	Anniversary / repeat-purchase rewards	Monthly	High	Family membership with bundled perks	Moderate referral credit
Budget-Conscious Families	Simple cashback or stamp-card style rewards	Value reinforcement messaging ('you saved \$X this year')	Bi-weekly	Standard	Free basic loyalty tier	Small discount-based referral incentive
Low-Engagement Shoppers	Low-barrier entry rewards to build habit	Re-engagement nudges tied to browsing behavior	Monthly (trigger-based)	Standard, self-serve first	No-cost membership to lower barrier to entry	Small one-time referral incentive

Task 8 — Customer Reactivation Strategy

Reactivation priority focuses on the Low-Engagement Shoppers segment (33% of the base, lowest spend and highest browse-to-buy ratio — they visit but rarely convert) and the subset of Budget-Conscious Families showing declining purchase activity.

Element	Recommendation
Customers needing reactivation	Low-Engagement Shoppers (highest priority); lower-frequency members of Budget-Conscious Families
Win-back offer	First-purchase-style discount (20-25%) or free gift with minimum spend
Reminder campaigns	Cart/browse abandonment emails; 'still interested?' nudges tied to web visit activity
Personalized recommendations	Best-seller and entry-level items matched to any prior browsing or purchase history
Reactivation timeline	Trigger at 30 days of inactivity (reminder), 60 days (offer), 90 days (final win-back + survey to learn why they disengaged)

Task 9 — Campaign Action Plan

Campaign	Target Segment	Objective	Channel	Budget Priority	Expected Outcome	KPI
VIP Early Access	Premium Loyal Customers	Increase loyalty & wallet share	Email + Catalog	Medium	Higher repeat purchase rate	Repeat Purchase Rate
Family Bundle Days	Affluent Family Shoppers	Increase basket size	Email + Web	High	Larger average order value	Average Order Value
Everyday Savings	Budget-Conscious Families	Convert browsing to buying	SMS + Email	Medium	Higher conversion from visits to purchases	Conversion Rate
Welcome Back	Low-Engagement Shoppers	Reduce churn / reactivate	Email + Retargeting	High	Reactivated customers, reduced inactivity	Reactivation Rate
Referral Rewards	Premium Loyal & Affluent Family	Acquire similar high-value customers	Email + Loyalty portal	Low	New customers from referrals	Referral Conversion Rate

Task 10 — Marketing Strategy Matrix (Consolidated)

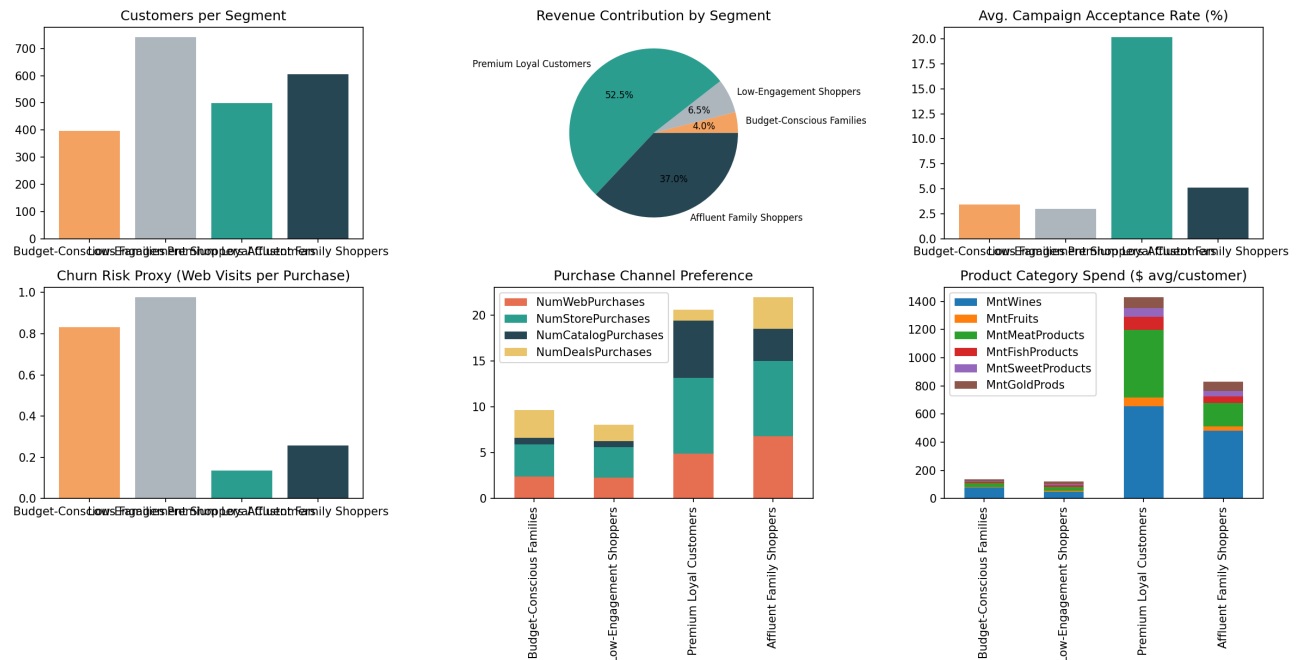
A single reference view combining message, channel, product focus, discount level, loyalty approach, cross-/up-sell direction, cadence, and reactivation posture for each segment.

Segment	Message	Channel	Product Focus	Discount	Loyalty Approach	Cross/Upsell	Frequency	Reactivation
Premium Loyal Customers	Exclusive access, first	Email + Catalog	Wine, meat, premium	0-5%	Invite-only VIP tier	Reserve wines / pairings	Weekly	Not applicable — retain
Affluent Family Shoppers	More of what your family loves	Email + Web	Wine, meat, family bundles	10-15%	Family points program	Larger packs / sides	Bi-weekly	Light — frequency nudges
Budget-Conscious Families	Save on the essentials	SMS + Email	Value wine, staples	15-20%	Free basic loyalty tier	Mid-tier upgrades	Bi-weekly	Moderate — conversion nudges
Low-Engagement Shoppers	We miss you, welcome back	Email + Retargeting	Best-sellers, entry-level	20-25%	No-cost entry membership	Starter bundles	Monthly	High — full win-back sequence

Task 11 — Business Insights Dashboard

Built in Python (pandas + matplotlib); see accompanying Jupyter notebook for reproducible code.

Customer Segmentation — Business Insights Dashboard



Reading the dashboard: Premium Loyal Customers and Affluent Family Shoppers together are 49% of customers by count but contribute roughly 90% of total revenue. Premium Loyal Customers have by far the highest campaign acceptance rate. The churn-risk proxy (web visits per completed purchase) is highest for Budget-Conscious Families and Low-Engagement Shoppers, both of which visit relatively often but convert far less than the two high-value segments.

Task 12 — Executive Management Summary

A standalone, board-ready version of this section is also provided as **Executive_Summary.pdf**. Below is the same content for completeness of this combined report.

Project Overview

This project segments 2,240 customers using demographic, spending, and engagement data, then translates the resulting four segments into marketing, pricing, retention, and reactivation recommendations to guide the next planning cycle.

Customer Segmentation Summary

K-Means clustering (k=4) on standardized behavioral and demographic features produced four segments: Premium Loyal Customers (22.2%), Affluent Family Shoppers (27.0%), Budget-Conscious Families (17.7%), and Low-Engagement Shoppers (33.1%).

Key Business Findings

Revenue is highly concentrated: the two higher-income segments (49% of customers) generate the large majority of total spend. Premium Loyal Customers respond to campaigns at roughly 4x the rate of any other segment. The largest segment by headcount, Low-Engagement Shoppers, contributes the least revenue per customer and shows the weakest conversion from browsing to buying.

High-Value Customer Segments

Premium Loyal Customers and Affluent Family Shoppers are the priority segments to protect and grow — through loyalty tiers, early access, and family-oriented bundles rather than heavy discounting, which would erode margin unnecessarily on customers who are already willing to pay.

High-Risk Customer Segments

Low-Engagement Shoppers carry the highest disengagement risk given low spend and low campaign response despite meaningful web activity. Budget-Conscious Families are price-sensitive and could churn to lower-cost alternatives if value messaging weakens.

Marketing Recommendations

Tailor message, channel, and offer type per segment (see Task 4/10): exclusivity-led messaging for high-value segments, value- and convenience-led messaging for price-sensitive and disengaged segments, with cadence matched to each group's natural engagement level.

Revenue Improvement Opportunities

Increasing basket size and frequency among Affluent Family Shoppers through bundling, and converting a portion of Budget-Conscious Families' browsing activity into purchases, represent the two largest near-term revenue levers alongside protecting Premium Loyal Customer wallet share.

Customer Retention Recommendations

Introduce a tiered loyalty program (see Task 7) so that Premium and Affluent segments have a clear reason to consolidate spend with the business, while low-cost, low-barrier loyalty mechanics help retain price-sensitive segments without over-discounting.

Business Growth Opportunities

Activating even a modest share of the Low-Engagement segment (33% of the base) through a structured reactivation sequence (Task 8) offers a meaningful volume opportunity given its size.

Final Conclusion

The segmentation shows a clear value hierarchy across the customer base. The recommended strategy is to protect and deepen relationships with the two high-value segments while running cost-efficient conversion and reactivation programs for the two lower-value segments, rather than applying one uniform marketing approach across all customers.

Final Deliverables Checklist

- Customer Segment Summary Table
- Customer Persona Profiles
- Business Opportunity Analysis
- Marketing Strategy Recommendations
- Product Recommendation Plan
- Pricing and Discount Strategy
- Customer Retention Strategy
- Customer Reactivation Plan
- Campaign Action Plan
- Marketing Strategy Matrix
- Business Insights Dashboard
- Executive Management Summary
- Well-documented Jupyter Notebook (.ipynb)
- Clean Python Source Code (.py)
- Final PDF Report with all findings